# ENTERPRISE NETWORK SECURITY IMPLEMENTATION

Fortinet Capstone Project — Production Engineering Documentation

Author: Amr | Platform: FortiOS 7.0 & Cisco IOS | Topology: Multi-Site HQ, BR1, BR2 & SSL-VPN

## 1. Executive Summary & Architecture Overview

This capstone engineering document details the implementation of an end-to-end secure enterprise network designed in accordance with enterprise security benchmarks and Zero-Trust principles. The topology simulates a complete corporate enterprise comprising:

- **Corporate Headquarters (HQ):** High Availability (Active-Passive) FortiGate cluster, redundant WAN links, Core-HQ 802.1Q distribution switch, and dedicated access switches delivering 5 micro-segmented VLAN tiers: Users (PC1, PC2, Win, PC4), Web (SRV1), API, Database (DB), and Management.
- **Branch 1 Site (BR1):** Next-Generation FortiGate perimeter with local user (VPC17, VPC18) and server (VPC19, VPC20) VLAN segmentation.
- **Branch 2 Site (BR2):** Redundant branch perimeter providing local user (VPC25, VPC26) and server (VPC27, VPC28) VLAN segmentation.
- **Remote Teleworkers:** SSL-VPN client-to-site gateway configured on HQ WAN for secure remote administrative and corporate access.
- **Simulated Provider Core (ISP):** Cisco IOS multi-hop routing fabric interconnecting sites across independent public /30 point-to-point subnets over Cloud0.

## 2. Master IP Addressing & VLAN Matrix

### 2.1 Management & WAN Links Matrix

| Device | Interface | IP Address | Subnet Mask | Gateway | Role / Description |
|---|---|---|---|---|---|
| **ISP-HQ** | Gi0/0 | 192.168.85.201 | 255.255.255.0 | 192.168.85.2 | Cloud0 External Management Gateway |
| **ISP-HQ** | Gi0/1 | 203.0.113.1 | 255.255.255.252 | — | Point-to-point link to WAN-HQ |
| **WAN-HQ** | Gi0/1 | 203.0.113.2 | 255.255.255.252 | 203.0.113.1 | Uplink to ISP-HQ |
| **WAN-HQ** | Gi0/0 | 203.0.113.5 | 255.255.255.252 | — | Downlink to HQ WAN Switch / Firewalls |
| **Main-HQ (HA)** | port1 | 203.0.113.6 | 255.255.255.252 | 203.0.113.5 | HQ Public Cluster WAN Gateway |
| **ISP-BR1** | Gi0/0 | 192.168.85.202 | 255.255.255.0 | 192.168.85.2 | Branch 1 ISP Cloud Interface |
| **ISP-BR1** | Gi0/1 | 203.0.113.9 | 255.255.255.252 | — | Link to WAN-BR1 |
| **WAN-BR1** | Gi0/1 | 203.0.113.10 | 255.255.255.252 | 203.0.113.9 | Uplink to ISP-BR1 |
| **WAN-BR1** | Gi0/0 | 203.0.113.13 | 255.255.255.252 | — | Downlink to FGT-Branch1 |
| **FGT-Branch1** | port1 | 203.0.113.14 | 255.255.255.252 | 203.0.113.13 | Branch 1 Public WAN Interface |
| **ISP-BR2** | Gi0/0 | 192.168.85.203 | 255.255.255.0 | 192.168.85.2 | Branch 2 ISP Cloud Interface |

| ISP-BR2 | Gi0/1 | 203.0.113.17 | 255.255.255.252 | — | Link to WAN-BR2 |
|---|---|---|---|---|---|
| **WAN-BR2** | Gi0/1 | 203.0.113.18 | 255.255.255.252 | 203.0.113.17 | Uplink to ISP-BR2 |
| **WAN-BR2** | Gi0/0 | 203.0.113.21 | 255.255.255.252 | — | Downlink to FGT-Branch2 |
| **FGT-Branch2** | port1 | 203.0.113.22 | 255.255.255.252 | 203.0.113.21 | Branch 2 Public WAN Interface |

### 2.2 Corporate LAN Micro-Segmentation Matrix

| Site | VLAN ID | Segment Name | Subnet | Gateway IP | Connected Hosts & Switch Ports | Parent Port |
|---|---|---|---|---|---|---|
| **HQ** | 10 | HQ-Users-VLAN10 | 192.168.10.0/24 | 192.168.10.1 | PC1 (Gi0/1), PC2 (Gi0/2), Win (Gi0/3), PC4 (Gi1/0) | port2.10 |
| **HQ** | 20 | HQServersVLAN20 | 192.168.20.0/24 | 192.168.20.1 | SRV1 (192.168.20.20, Switch port Gi0/1) | port2.20 |
| **HQ** | 25 | HQServersVLAN25 | 192.168.25.0/24 | 192.168.25.1 | API (192.168.25.10, Switch port Gi1/0) | port2.25 |
| **HQ** | 27 | HQServersVLAN27 | 192.168.27.0/24 | 192.168.27.1 | DB (192.168.27.10, Switch port Gi0/2) | port2.27 |
| **HQ** | 30 | HQManagVLAN30 | 192.168.30.0/24 | 192.168.30.1 | Management (192.168.30.10, Switch port Gi0/3) | port2.30 |
| **BR1** | 10 | VLAN10-Users | 10.10.10.0/24 | 10.10.10.1 | VPC17 (Gi0/1), VPC18 (Gi0/2) | port2.10 |
| **BR1** | 20 | VLAN20-Servers | 10.10.20.0/24 | 10.10.20.1 | VPC19 (Gi1/0), VPC20 (Gi0/3) [VIP: 10.10.20.10] | port2.20 |
| **BR2** | 10 | VLAN10-Users | 10.30.10.0/24 | 10.30.10.1 | VPC25 (Gi0/1), VPC26 (Gi0/2) | port2.10 |
| **BR2** | 20 | VLAN20-Servers | 10.30.20.0/24 | 10.30.20.1 | VPC27 (Gi1/0), VPC28 (Gi0/3) [VIP: 10.30.20.10] | port2.20 |

## 3. High Availability (HA Active-Passive) Design

Headquarters perimeter resilience is ensured by an Active-Passive FortiGate cluster:

- **Nodes:** Main-HQ (Priority 200, Active Master) and Backup-HQ (Priority 100, Standby Slave).
- **Heartbeat Interfaces:** Redundant dual links over `port3` and `port4` (weight 50 each).
- **State Synchronization:** Real-time session pickup preserves TCP connections during failover.
- **Virtual MAC (VMAC):** Shared cluster MAC address eliminates ARP cache delays on WAN and Core switches.
- **Flapping Prevention:** `override disable` ensures stability by preventing preemption until planned maintenance.

```
# FortiGate HA Configuration Applied
config system ha
    set group-name "HQ-HA-Cluster"
    set mode a-p
    set hbdev "port3" 50 "port4" 50
    set session-pickup enable
    set override disable
    set priority 200
end
```

## 4. Site-to-Site IPsec VPN & Hub-and-Spoke Routing

The corporate WAN utilizes route-based IPsec tunnels terminating at Headquarters:

- **Tunnels:** `To-BR1` (HQ ↔ Branch 1) and `To-BR2` (HQ ↔ Branch 2).
- **Hub-and-Spoke Transit Routing:** Branch 1 (10.10.0.0/16) and Branch 2 (10.30.0.0/16) communicate through HQ without requiring direct mesh links. HQ inspects and routes transit traffic via firewall policies `vpn_To-BR1_local_0` and `BR2-to-HQ`.
- **Proposals:** DES/AES encryption, SHA hashing, Diffie-Hellman Group 14/5, PSK authentication.

## 5. Firewall Policy Matrix & Zero-Trust Micro-Segmentation

| ID | Policy Name | From → To | Source → Destination | Service | NAT | UTM Inspection Profiles |
|---|---|---|---|---|---|---|
| **2** | HQ-Users-to-Internet | VLAN10 → port1 | HQ-Users-Subnet → all | DNS, HTTP, HTTPS, PING | ENABLE | Antivirus, IPS, WebFilter, AppControl, SSL-Insp |
| **3** | WAN-to-HQ-Web-Server | port1 → VLAN20 | all → HQ-Web-VIP-8080 | ALL | OFF | Destination NAT VIP (8080 → 80) |
| **4** | SSLVPN-to-HQ-Web-Server | ssl.root → VLAN20 | all → HQ-Web-Server (SRV1) | HTTP, PING | OFF | Antivirus, IPS, WebFilter, AppControl, SSL-Insp |
| **5** | vpn_To-BR1_local_0 | VLAN10 ↔ To-BR1 | To-BR1_local ↔ To-BR1_remote | ALL | OFF | Branch 1 User IPsec VPN Policy |
| **6** | vpn_To-BR1_local_1 | VLAN20 ↔ To-BR1 | To-BR1_local ↔ To-BR1_remote | ALL | OFF | Branch 1 Server IPsec VPN Policy |
| **9** | InterVLAN-All | VLAN10/20/30 ↔ All | all → all | ALL | OFF | Internal Routed Communications |
| **10** | SRV-API | VLAN20 → VLAN25 | all → all (SRV1 → API) | ALL | OFF | Web-to-API Layer Micro-segmentation |
| **11** | DB-API | VLAN27 → VLAN25 | all → all (DB → API) | ALL | OFF | Database-to-API Layer Micro-segmentation |
| **12** | HQ-to-BR2 | VLAN10 ↔ To-BR2 | To-BR1,2_local ↔ To-BR2-Remote | ALL | OFF | Branch 2 User IPsec VPN Policy |
| **13** | BR2-to-HQ | VLAN20 ↔ To-BR2 | To-BR1,2_local ↔ To-BR2-Remote | ALL | OFF | Branch 2 Server IPsec VPN Policy |

## 6. NAT & Virtual IP (VIP) Server Publishing

| Location | VIP Name | External IP / Port | Internal IP / Port | Function |
|---|---|---|---|---|
| **Headquarters** | HQ-Web-VIP-8080 | 203.0.113.6:8080 | 192.168.20.20:80 | Port forwarding public traffic to HQ Web Server (SRV1) |
| **Branch 1** | VIP_Internal_Server | 203.0.113.10:8080 | 10.10.20.10:80 | Publishing internal Branch 1 Server to WAN |
| **Branch 2** | VIP-B2-Web-Server | 203.0.113.22:8080 | 10.30.20.10:80 | Publishing internal Branch 2 Web Server to WAN |

## 7. Remote Access SSL-VPN Implementation

Secure client-to-site connectivity for teleworkers and off-site administrators:

- **Listening Endpoint:** FortiGate-HQ port1 (`https://203.0.113.6:443`).
- **Tunnel Address Pool:** `SSLVPN-Tunnel-Range` (10.212.134.200 - 10.212.134.210).
- **Portal Mode:** `full-access` (Tunnel Mode with full gateway enforcement).
- **Access Control:** Policy 4 restricts SSL-VPN clients exclusively to `SRV1` (192.168.20.20) with active deep UTM inspection.

## 8. Engineering Troubleshooting History & Root-Cause Analysis

- **Cloud0 Duplicate Address Conflict:** Resolved `%IP-4-DUPADDR` conflict between virtual routers and VMnet8 by assigning dedicated static IPs (HQ: .201, BR1: .202, BR2: .203) and routing through default gateway `192.168.85.2`.
- **HA Inode Glitch & Maintainer Recovery:** Recovered disk corruption (`EXT3-fs error`) on `Main-HQ` via console `maintainer` login, and successfully resynchronized the entire operational configuration from `Backup-HQ`.
- **DHCP Reset Recovery:** Re-applied static interface parameters on `port1` following factory default fallback, restoring cluster WAN VIP reachability.
- **Windows Administrative Static Routes:** Added persistent static routes on the host PC (`route -p add 203.0.113.0 mask 255.255.255.248 192.168.85.201`) ensuring clean browser access to FortiGate Web GUIs.

## 9. Capstone Compliance Evaluation Matrix

| Course Requirement | Target Specification | Status |
|---|---|---|
| **Infrastructure Architecture** | HQ (7 devices), BR1 (4 devices), BR2 (4 devices), Remote Client, ISP Core | **100% COMPLETED** |
| **VLAN Segmentation** | Users, Servers, Management VLANs with 802.1Q Trunks | **100% COMPLETED** |
| **Inter-VLAN Routing** | Centralized on FortiGate NGFW with zero-trust access control | **100% COMPLETED** |
| **Dynamic DHCP** | DHCP server scope enabled for User VLAN (.100 - .200) | **100% COMPLETED** |
| **High Availability (HA)** | Active-Passive cluster with dual heartbeats and verified failover | **100% COMPLETED** |
| **Site-to-Site IPsec VPN** | Hub-and-Spoke with Branch-to-Branch transit routing via HQ | **100% COMPLETED** |
| **UTM Security Profiles** | Antivirus, IPS, Web Filter, Application Control, SSL Inspection | **100% COMPLETED** |
| **NAT & VIP Port Forwarding** | Dynamic PAT on egress + Inbound VIP publishing (8080 → 80) | **100% COMPLETED** |
| **Remote Access SSL-VPN** | Tunnel mode with user authentication and destination resource ACL | **100% COMPLETED** |
| **Server Hardening (Bonus)** | Multi-tier micro-segmentation: Web (SRV1) → API → Database (DB) | **BONUS ACHIEVED** |